

# Thiago Ferreira dos Santos

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## RESEARCH INTERESTS

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**Stellar Astrophysics.** Pop. III, stellar abundances, oscillations, activity, white dwarfs, wide and close binary systems;  
**Exoplanets and Brown Dwarfs.** evolution and star-planet interactions.  
**Galactic Structure and Archaeology.** stellar surveys, Galactic bulge and disc, stellar content, and infrared observations;

## EDUCATION

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|-------------------------------------------------------------------------------|----------------|
| <b>PhD candidate in Astrophysics – Yale University, US</b>                    | 2023 – Present |
| <b>MSc in Astronomy – Universidade de São Paulo, Brazil</b>                   | 2021 – 2023    |
| <b>BSc in Physics – Universidade Federal de Santa Catarina (UFSC), Brazil</b> | 2016 – 2021    |

## REFERRED PUBLICATIONS

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As of **May 2026**: **14** peer-reviewed (**6** as first-author) | **24** total | h-index: **7** | Citations: **151** | [ADS Library](#) | [Google Scholar](#)

14. **Ferreira, T.**, Bellinger, E.P., Farag, E., Lindsay, C.J., [ApJ](#), 1001, 41 (2026)

*Red-Giant Asteroseismology of Low-Mass Population III Stars*

13. **Ferreira, T.**, Bellinger, E.P., Farag, E., Lindsay, C.J., [ApJ](#), 1000, 147 (2026)

*Evolution of Low-Mass Population III Stars: Convection, Mass Loss, Nucleosynthesis, and Neutrinos*

12. **Ferreira, T.**, Yana Galarza, J., et al., [accepted for publication in AJ](#) (2026)

*Hidden Companions of the Early Milky Way I. New  $\alpha$ -Enhanced Exoplanet Hosts*

11. Winnick, I., et al. (incl. **Ferreira, T.**), [accepted for publication in AJ](#) (2025)

*Extreme Lithium Depletion in Solar Twins: Challenging Non-Standard Mixing Models*

10. Galarza, J.Y., Lorenzo-Oliveira, D., **Ferreira, T.**, et al., [ApJ](#), 983, 70 (2025)

*HIP 8522: A Puzzling Young Solar Twin with the Lowest Detected Lithium*

9. Fermiano, V., et al. (incl. **Ferreira, T.**), [A&A](#), 690, L7 (2024)

*The Young Exoplanetary System TOI-4562: Confirming The Presence of a Third Body in the System*

8. **Ferreira, T.**, Rice, M., et al., [AJ](#), 168, 145 (2024)

*SOLES XI. The Aligned Orbit of TOI-2533 b, a Transiting Brown Dwarf Orbiting an F8-type Star*

7. Galarza, J.Y., **Ferreira, T.**, Lorenzo-Oliveira, D., et al., [AJ](#), 168, 91 (2024)

*TOI-1173 A b: The First Inflated Super-Neptune in a Wide Binary System*

6. Galarza, J.Y., Reggiani, H., **Ferreira, T.**, et al., [ApJ](#), 974, 122 (2024)

*Detailed Abundances of the Planet-Hosting TOI-1173 A/B: Possible Evidence of Planet Engulfment in a Very Wide Binary*

5. Saito, R.K., et al. (incl. **Ferreira, T.**), [A&A](#), 689, A148 (2024)

*The VISTA Variables in the Vía Láctea eXtended (VVVX) ESO Public Survey: Completion of the Observations and Legacy*

4. **Ferreira, T.**, Saito, R.K., Minniti, D., et al., [MNRAS](#), 527, 10737 (2024)

*A Benchmark White Dwarf–Ultracool Dwarf Wide Field Binary*

3. Botan, E., et al. (incl. **Ferreira, T.**), [MNRAS](#), 504, 654 (2021)

*Unveiling Short-Period Binaries in the Inner VVV Bulge*

2. Herpich, F.R., et al. (incl. **Ferreira, T.**), [A&A](#), 647, A169 (2021)

*VVV Survey Near-Infrared Colour Catalogue of Known Variable Stars*

1. **Ferreira, T.**, Saito, R.K., Minniti, D., et al., *MNRAS*, 486, 1220 (2019)  
*The Asymptotic Evolution of the Stellar Merger V1309 Sco: a Blue Straggler in the Making?*

## TELESCOPE OBSERVING EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                      |              |
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| <b>Keck I 10-m / HIRES</b> – 2 nights, PI<br><i>Uncovering Early Supernova Signatures in CEMP Halo Giant Stars</i>                                                                                                                                                                                                                                   | 2026A        |
| <b>Subaru 8.2-m / HDS</b> – 1.5 nights, PI<br><i>Uncovering Early Supernova Signatures in CEMP Giant Stars</i>                                                                                                                                                                                                                                       | 2026A        |
| <b>Gemini South 8.1-m / GHOST</b> – 1.5 nights, PI<br><i>Uncovering Early Supernova Signatures in CEMP Giant Stars</i>                                                                                                                                                                                                                               | 2026A        |
| <b>Magellan Clay 6.5-m / MIKE &amp; PFS</b> – 2 nights, Co-I<br><i>Exploring the Origin of Exoplanets in the Milky Way's Thick-Disk</i>                                                                                                                                                                                                              | 2026A        |
| <b>Gemini North/South 8.1-m / GHOST &amp; MAROON-X</b> – 2 nights, Co-I<br><i>Exploring the Origin of Exoplanets in the Milky Way's Thick-Disk</i>                                                                                                                                                                                                   | 2026A        |
| <b>Gemini South 8.1-m / GHOST</b> – 0.5 nights, DDT                                                                                                                                                                                                                                                                                                  | 2025B        |
| <b>Magellan Clay 6.5-m / WINERED</b> – 1 night, PI<br><i>Investigating Atmospheric Outflows in a Neptunian-desert Planet Using He I <math>\lambda 10\,833\text{ \AA}</math></i>                                                                                                                                                                      | 2025A        |
| <b>Magellan Clay 6.5-m / PFS</b> – 2 nights, Co-I<br><i>Confirming Thick Disc Membership for Known Exoplanets and TESS objects of interest</i>                                                                                                                                                                                                       | 2025A        |
| <b>Magellan Clay 6.5-m / MIKE</b> – 12 nights, Co-I<br><i>Exploring the star-planet connection with solar-type hosts</i><br><i>Correlated Abundances of Exoplanets and Their Host Stars: Constraints from Polluted White Dwarfs in Stellar Binaries</i><br><i>Confirming Thick Disc Membership for Known Exoplanets and TESS objects of interest</i> | 2024B, 2025A |
| <b>ESO 3.6-m / HARPS</b> – 6 nights, PI<br><i>How rare are analogues of the Solar System?</i>                                                                                                                                                                                                                                                        | 2023A (P111) |

## SEMINARS

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|-------------------------------------------------------------------------------------------------|----------|
| <b>Asteroseismology of Low-Mass Population III Stars</b> – TOS group, HITS/Heidelberg (Germany) | Jul 2025 |
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## CONFERENCE TALKS & POSTERS

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| 9th TESS/16th Kepler Asteroseismic Science Consortium Workshop, Vienna (poster)<br><i>On the Structure and Oscillations of the First Stars.</i>                | Jul 2025 |
| exoLaTam 22, Santiago (talk)<br><i>HIP 102152b: A Low-Mass Planet Candidate Around an Old Solar Twin.</i>                                                      | Dec 2022 |
| XLV Annual Meeting of the Brazilian Astronomical Society, Virtual (talk & poster)<br><i>HIP 102152b: A Low-Mass Planet Candidate Around an Old Solar Twin.</i> | Oct 2022 |
| Cool Stars 21, Virtual (poster)<br><i>HIP 102152b: A Low-Mass Planet Candidate Around an Old Solar Twin.</i>                                                   | Jul 2022 |
| Exoplanets III, Virtual (poster)<br><i>A Box-Fitting Implementation for the Transit Method Detection of Exoplanets.</i>                                        | Jul 2021 |
| Sagan Exoplanet Summer Workshop, Virtual (poster)<br><i>Time-Series Analysis of Near-IR Variabilities on VVV M Dwarf Stars.</i>                                | Jul 2021 |
| ALMA Planets2020 Workshop, Santiago (poster)                                                                                                                   | Mar 2020 |

*Exoplanetary Transits in the Galactic Bulge.*

EWASS 2019, Lyon (poster) Jun 2019

*Infrared Counterparts of Gaia Alerts (Ivanov+*

VVV & VVV-X Science Workshop, Florianópolis (poster) Apr 2018

*On the Photometric Study of the Cataclysmic Variable IGR J17014-4306.*

Seminário de Iniciação Científica, UFSC, Virtual (talk) 2018, 2019, 2020, 2021

*Stellar Variability Towards the Central Region of the Milky Way with the VVV/VVV-X Surveys.*

PPGFSC/UFSC Meeting, Florianópolis (poster) 2018, 2020

*Stellar Variability Towards the Central Region of the Milky Way with the VVV/VVV-X Surveys.*

## FELLOWSHIPS, AWARDS & GRANTS

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Yale Conference Travel Fund - Newton Endowment Jul 2025

GSAS Yale Graduate Fellowship 2023 – Present

Boris Garfinkel Prize Fellowship in Astronomy 2023

CAPES M.Sc. Research Fellowship, IAG/USP, Brazil 2021 – 2023

exoLaTam 2022 Travel Award Dec 2022

Teaching Improvement Program, IAG/USP 2022

ESO/ALMA Planets2020 Workshop Travel Award Mar 2020

CNPq Undergraduate Research Fellowship, UFSC, Brazil 2017 – 2021

## WORKSHOPS & ADVANCED SCHOOLS

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YCIU Primordial Black Holes Workshop, US Mar 2026

Yale AstroCodEx Hack Day, New Haven Aug 2025

MESA Summer School, Leuven, Belgium Jul 2025

Yale Tinsley Workshop, New Haven Oct 2024

New York Area Exoplanets Meeting, New York May 2024

ESO Star–Planet Connection Workshop, Virtual Oct 2021

XIII CBPF School, Virtual Aug 2021

ZTF Summer School on Variable Stars, Virtual Aug 2021

Sagan Exoplanet Summer Workshop, Virtual Jul 2021

## SERVICE AND TEACHING

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Referee – ApJ 2024 – Present

Teaching Assistant, ASTR 110, 220, 310, 556 – Yale University Aug 2024 – May 2026

Teaching Assistant, AGA 0502 – IAG/USP 2022

## ACADEMIC REFERENCES

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**Earl P. Bellinger** (PhD advisor) – Yale University

[earl.bellinger@yale.edu](mailto:earl.bellinger@yale.edu)

**Jorge L. Meléndez** (MSc advisor) – Universidade de São Paulo

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**Roberto K. Saito** (BSc advisor) – Universidade Federal de Santa Catarina

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